```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
import os
import urllib.request

print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

```
image_path = "/content/African_Lions.jpg"

# Load in color (BGR by default in OpenCV)
img_bgr = cv2.imread(image_path)

if img_bgr is None:
    print("Error: Image not found at", image_path
else:
    img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR
    plt.figure(figsize=(6,6))
    plt.imshow(img_rgb)
    plt.title("Original Image")
    plt.axis("off")
    plt.show()
```

Original Image

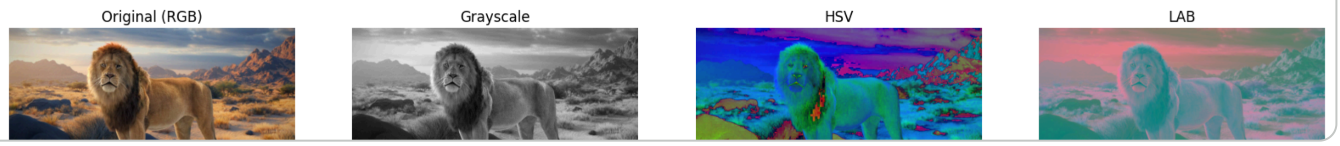
```
print("Image Shape (H, W, Channels):", img_bgr.sh
print("Image Size (total pixels):", img_bgr.size)
print("Image Data Type:", img_bgr.dtype)
print("Height:", img_bgr.shape[0])
print("Width:", img_bgr.shape[1])
print("Number of Channels:", img_bgr.shape[2] if
```

```
Image Shape (H, W, Channels): (1080, 1920, 3)
Image Size (total pixels): 6220800
Image Data Type: uint8
Height: 1080
Width: 1920
Number of Channels: 3
```

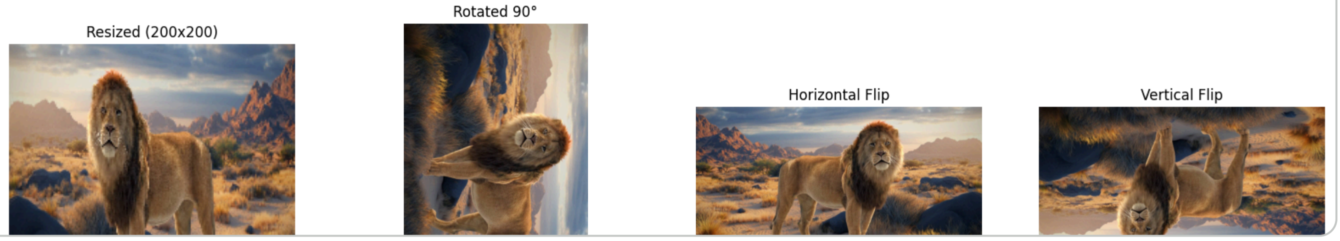
```
cv2.imwrite("output.jpg", img_bgr)
cv2.imwrite("output.png", img_bgr)

import os
print("JPEG size:", os.path.getsize("output.jpg") / 1024, "KB")
print("PNG size:", os.path.getsize("output.png") / 1024, "KB")
```

```
JPEG size: 347.84765625 KB
PNG size: 2382.2158203125 KB
```

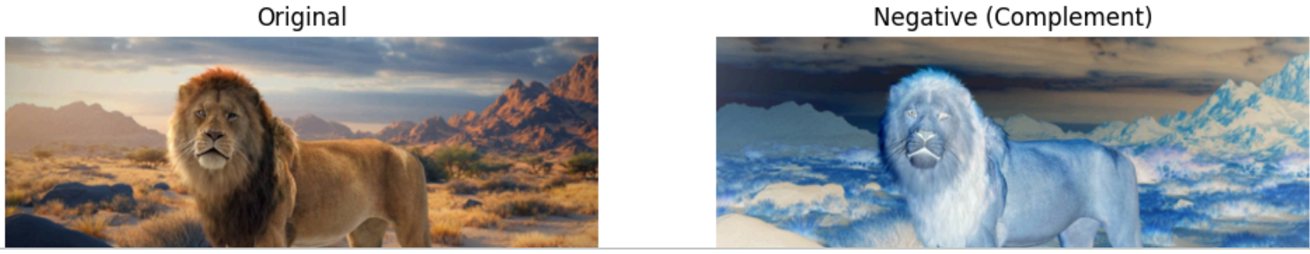
```
gray = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2GRAY)
hsv = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2HSV)
lab = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2LAB)

fig, axs = plt.subplots(1, 4, figsize=(20,5))
axs[0].imshow(img_rgb); axs[0].set_title("Original (RGB)"); axs[0].axis("off")
axs[1].imshow(gray, cmap="gray"); axs[1].set_title("Grayscale"); axs[1].axis("off")
axs[2].imshow(hsv); axs[2].set_title("HSV"); axs[2].axis("off")
axs[3].imshow(lab); axs[3].set_title("LAB"); axs[3].axis("off")
plt.show()
```



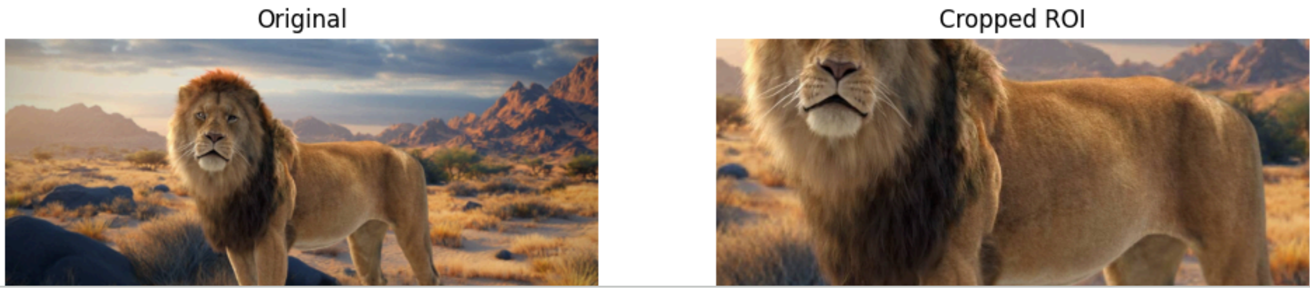

```
# Resize
resized = cv2.resize(img_rgb, (200, 200))

# Rotate 90 degrees
rotated = cv2.rotate(img_rgb, cv2.ROTATE_90_CLOCKWISE)

# Horizontal flip
h_flip = cv2.flip(img_rgb, 1)

# Vertical flip
v_flip = cv2.flip(img_rgb, 0)

fig, axs = plt.subplots(1, 4, figsize=(20,5))
axs[0].imshow(resized); axs[0].set_title("Resized (200x200)"); axs[0].axis("off")
axs[1].imshow(rotated); axs[1].set_title("Rotated 90°"); axs[1].axis("off")
axs[2].imshow(h_flip); axs[2].set_title("Horizontal Flip"); axs[2].axis("off")
axs[3].imshow(v_flip); axs[3].set_title("Vertical Flip"); axs[3].axis("off")
plt.show()
```




```
negative = 255 - img_rgb

fig, axs = plt.subplots(1, 2, figsize=(12,6))
axs[0].imshow(img_rgb); axs[0].set_title("Original"); axs[0].axis("off")
axs[1].imshow(negative); axs[1].set_title("Negative (Complement)"); axs[1].axis("off")
plt.show()
```




```
h, w = img_rgb.shape[:2]
y1, y2 = int(h*0.25), int(h*0.75)
x1, x2 = int(w*0.25), int(w*0.75)

roi = img_rgb[y1:y2, x1:x2]

fig, axs = plt.subplots(1, 2, figsize=(12,6))
axs[0].imshow(img_rgb); axs[0].set_title("Original"); axs[0].axis("off")
axs[1].imshow(roi); axs[1].set_title("Cropped ROI"); axs[1].axis("off")
plt.show()

print("ROI Shape:", roi.shape)
```




```
fig, axs = plt.subplots(2, 4, figsize=(20,10))

axs[0,0].imshow(img_rgb); axs[0,0].set_title("Ori
axs[0,1].imshow(gray, cmap="gray"); axs[0,1].set_
axs[0,2].imshow(hsv); axs[0,2].set_title("HSV"); 
axs[0,3].imshow(lab); axs[0,3].set_title("LAB"); 

axs[1,0].imshow(rotated); axs[1,0].set_title("Rot
axs[1,1].imshow(h_flip); axs[1,1].set_title("H-Fl
axs[1,2].imshow(negative); axs[1,2].set_title("Ne
axs[1,3].imshow(roi); axs[1,3].set_title("Cropped

plt.tight_layout()
plt.show()
```